STAC AI BOOTCAMP NYC




# Evaluation and evidence attribution

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# The decision this lecture supports

Is this system ready for this task—and what should we fix first?

- A benchmark score estimates performance on its tested cases.
- A diagnostic test isolates a specific capability or failure.
- An evidence trace tells us which sources and operations support a claim.

Source: Choi, Stanford CS224N Lecture 11 (2026), pp.15–26; HELM (Liang et al., 2022)

# Roadmap: from a claim to a defensible test

| Stage | Question | Time |
|---|---|---|
| 1. Define | What exactly must work? | 10 min |
| 2. Measure | Which observations establish success? | 10 min |
| 3. Intervene | What changes should matter? | 15 min |
| 4. Diagnose | Which dependency explains the failure? | 15 min |
| 5. Decide | What can we claim and release? | 10 min |

Source: Workshop synthesis / original teaching example

# Five learning objectives

- Write a task contract with observable success and abstention conditions.
- Choose separate measures of correctness, support, stability and calibration.
- Design paired interventions with repeat controls and uncertainty estimates.
- Distinguish evidence dependencies from causal effects inside a model.
- Turn failures into a repair, retest and bounded release claim.

Source: Workshop synthesis / original teaching example

# Start with an evaluation contract

| Specify before testing | Why it is needed |
|---|---|
| Input and allowed evidence | Defines what the system may know. |
| Output and permitted abstention | Makes success observable, including missing information. |
| Target cases and exclusions | Defines where the result applies. |
| Errors, costs and release tolerance | Determines which failures matter operationally. |

Source: Choi, Stanford CS224N Lecture 11 (2026), pp.15–26, 54–55 (adapted)

# A benchmark is a sample of the task

- Cover ordinary cases, difficult cases and cases that cannot be answered.
- Include shortcuts that work on easy cases but fail on carefully matched cases.
- Keep independent test families separate from development examples.
- Report results by failure type as well as overall.

Source: Choi, Stanford CS224N Lecture 11 (2026), pp.15–26

# A labelled evaluation record

| Field | Illustrative record |
|---|---|
| Case and family | E01; meeting-update family. |
| Input | The meeting moved from Tuesday to Friday. Which day now? |
| Reviewed reference | Friday. |
| Support label | Friday is supported; Tuesday is superseded. |
| Missing-source variant | Expected behavior: withhold the unsupported answer. |

Keep case families separate from development. Do not train on the final evaluation fixtures.

# Correctness, support and stability are different

| Property | Question | Possible failure |
|---|---|---|
| Correctness | Does it match the reference? | Correct answer by lucky guessing. |
| Support | Do eligible sources justify it? | Plausible claim with irrelevant citation. |
| Stability | Does irrelevant variation change it? | Same facts, different answer order. |
| Calibration | Does stated probability match frequency? | Confident errors on unseen cases. |

Source: Liang et al., HELM (2022); Guo et al., Calibration (ICML 2017)

# Choose the measurement for the claim

| Target | Observable | What it does not establish |
|---|---|---|
| Exact extraction / arithmetic | Exact match; recomputed value | Whether the source is eligible. |
| Free-text support | Atomic claim + cited span + rubric | Whether the model relied on it. |
| Operational behavior | Executed tool and action trace | Whether the answer is true. |
| Language similarity | Overlap or embedding similarity | Entailment, factuality or authorization. |

Source: Choi, Stanford CS224N Lecture 11 (2026), pp.34–41, 51–55 (adapted)

# Evaluate the evaluator

- Write a rubric with correct, unsupported, contradicted and unanswerable examples.
- Have independent reviewers label a shared sample; inspect disagreement.
- If a model assists grading, validate it against reviewed labels and retain reasons.
- Use accountable checks and review for release decisions.

Source: Choi, Stanford CS224N Lecture 11 (2026), pp.54–64

# Question: a correct answer cites the right document

## What have we established?

- A. The model must have used that document.
- B. Correctness and a citation are observed; dependence still needs testing.
- C. The answer will remain correct after the document changes.

Source: Workshop synthesis / original teaching example

# Answer: B—test dependence separately

- Check that the cited passage actually supports the claim.
- Change or remove that evidence under a defined intervention.
- Compare the outcome with a matched control.
- Interpret the result only for the tested intervention.

Source: DeYoung et al., ERASER (ACL 2020)

# Three kinds of behavioral test

| Test | What should happen? | Simple example |
|---|---|---|
| Minimum functionality | A basic capability works. | Extract the stated date. |
| Invariance | Meaning-preserving edits preserve the result. | Change the order of independent records. |
| Directional expectation | Relevant changes move the result appropriately. | Replace positive evidence with a contradiction. |

Source: Ribeiro et al., CheckList (ACL 2020)

# Control one difference at a time

| Arm | Held fixed | Changed |
|---|---|---|
| Baseline repeats | Evidence, wording, order, settings | Execution only. |
| Presentation intervention | Evidence content and task | Framing or record order. |
| Evidence intervention | Task and presentation template | One relevant fact or missing source. |
| Negative control | Relevant evidence and task | An unrelated passage. |

Source: Ribeiro et al., CheckList (ACL 2020); Choi, Stanford CS224N Lecture 11 (2026), p.76

# Measure changes, not just final labels

$$\text{flip rate} = \frac{1}{n}\sum_i \mathbf{1}\{a_i \neq a_i'\} \qquad \text{TV}(q_i, q_i') = \frac{1}{2}\sum_a |q_i(a) - q_i'(a)|$$

- Flip rate detects changed selected actions.
- Total variation detects probability movement even when the top action stays fixed.
- Keep correctness alongside sensitivity: an unchanged wrong answer is still wrong.

Source: Workshop synthesis / original teaching example

# Candidate scoring requires a fixed boundary

$$\ell(a \mid x) = \sum_{t=1}^{T_a} \log p(a_t \mid x, a_{<t}), \qquad q(a \mid x) = \frac{e^{\ell(a|x)}}{\Sigma_{b \in \mathcal{A}} e^{\ell(b|x)}}$$

- Score every candidate under the same prompt and answer prefix.
- Record tokenization, termination and any length normalization.
- Missing alternatives from an API top-k response are unknown, not zero.

Source: Workshop synthesis / original teaching example

# Calibration needs outcomes and a separate split

$$\Pr(Y = 1 \mid \hat{p} \approx 0.8) \approx 0.8$$

- Define the event: e.g. this extracted field is correct.
- Fit any calibrator on reviewed calibration cases.
- Evaluate reliability on untouched cases from the intended population.
- Check performance after changes in sources, prompts or model.

Source: Guo et al., Calibration (ICML 2017)

# Statistical evidence must match the test design

| Comparison | Approach | Unit that must remain paired |
|---|---|---|
| Binary correctness, two systems | McNemar test; discordant cases | Same case under both systems. |
| Mean score difference | Paired bootstrap interval | Independent case or case family. |
| Many permutations per case | Summarize within case first | Do not count permutations as new cases. |

Source: Dror et al., Statistical Significance in NLP (ACL 2018)

# A small p-value does not establish safe use

- Report effect size and an uncertainty interval.
- Set a practically acceptable sensitivity tolerance before looking at results.
- Failure to reject zero difference does not prove equivalence.
- Account for multiple tested prompts, metrics and model comparisons.

Source: Dror et al., Statistical Significance in NLP (ACL 2018)

# Class Exercise: specify a controlled evaluation

| Step | Write down |
|---|---|
| 1. Choose a claim | The answer should follow the supplied evidence. |
| 2. Define three edits | Paraphrase; relevant contradiction; unrelated deletion. |
| 3. Add a baseline | Repeat the identical prompt and exact evidence order. |
| 4. Specify observations | Correctness, action flips and any available candidate scores. |

Source: Workshop synthesis / original teaching example

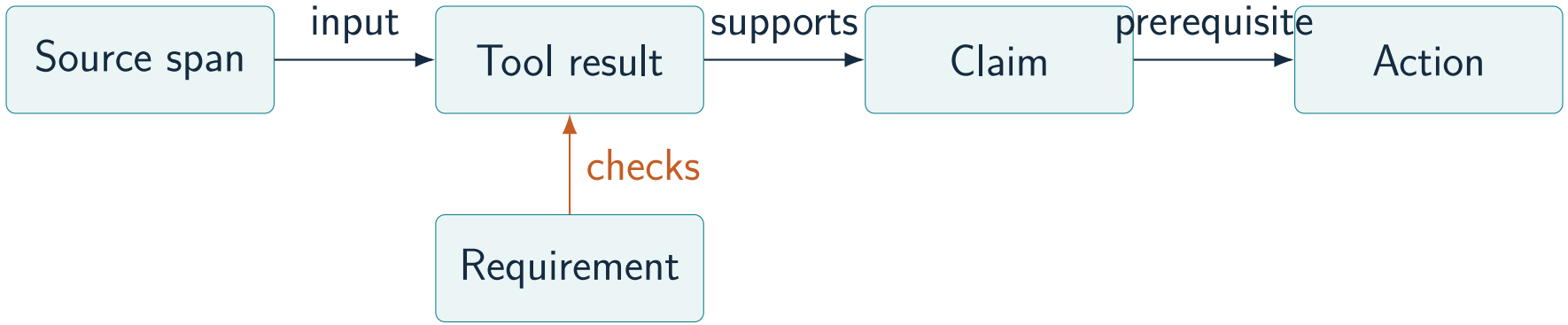
# Why inspect dependencies after scoring?

A failure rate tells us how often. A dependency trace helps identify what to repair.

| Recorded object | Diagnostic question |
|---|---|
| Source span | Is this the fact the answer claims to use? |
| Tool input and output | Were the cited values actually passed to the calculator? |
| Derived claim | Which assumptions and prerequisites support it? |
| Proposed action | Which unresolved claims and permissions matter? |

Source: Workshop synthesis / original teaching example

# An evidence graph makes the repair path visible




- Record edge meanings and provenance; retain disputed links.
- A missing input marks downstream support unresolved. Recheck after repair.

Source: Workshop synthesis / original teaching example

# Support, dependence and mechanism form a ladder

| Claim | Evidence needed |
|---|---|
| The passage supports the answer | Reviewed entailment and eligibility. |
| Changing the passage changes the answer | Controlled input intervention. |
| A particular hidden state mediates the effect | Internal intervention with controls. |
| The repair improves deployment outcomes | Independent outcome evaluation. |

Source: DeYoung et al., ERASER (ACL 2020); Meng et al., Causal Tracing / ROME (NeurIPS 2022)

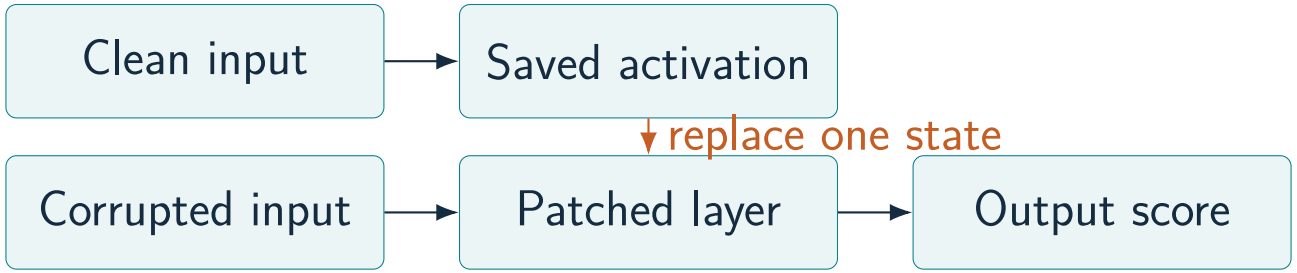
# Remove evidence; then retain only that evidence

$$\text{Comp} = p_y(x) - p_y(x \setminus R), \qquad \text{Suff} = p_y(x) - p_y(R)$$

- Comprehensiveness: how much does removing rationale R reduce the score?
- Sufficiency: how much score is lost when only R remains?
- Compare with matched unrelated removals; redundant evidence can hide effects.

Source: DeYoung et al., ERASER (ACL 2020)

# Patching asks a narrower internal question




$$\Delta = s(\text{corrupted run with patch}) - s(\text{corrupted run})$$

Source: Meng et al., Causal Tracing / ROME (NeurIPS 2022)

# Use the graph to select tests, then verify them

- Start from a failed claim and identify its source spans and tool inputs.
- Test a source-level repair before intervening inside the model.
- If needed, map the changed text to token positions for a controlled patch.
- Retest the repaired workflow on independent cases.

Source: Workshop synthesis / original teaching example

# Question: no answer changes across reorderings

## Which conclusion is warranted?

- A. The model is reliable in general.
- B. No selected-answer changes occurred in these tested cases.
- C. Its candidate probabilities must also be unchanged.

Source: Workshop synthesis / original teaching example

# Answer: B—state the tested boundary

- Report the cases, permutations, repeats and output parser.
- Measure probability movement separately when complete scores are available.
- Check correctness: stable errors remain errors.
- More general reliability requires broader cases and an explicit tolerance.

Source: Workshop synthesis / original teaching example

# Recorded pilot: low sensitivity can coexist with errors

| Observed in the expanded framing pilot | Interpretation |
|---|---|
| 1,200 assessments across two models | A controlled synthetic evaluation, not population prevalence. |
| Bullish framing did not increase unsupported buys | No evidence for that specific directional effect in this test. |
| Qwen mishandled some evidence gaps; Llama passed the benchmark | Correctness and framing sensitivity must be reported separately. |

Source: Workshop recorded expanded-framing experiment; original run artifacts retained

# What BeliefLens adds to this evaluation loop

| Activity | Concrete output |
| --- | --- |
| Organize evidence and executed steps | Traceable nodes, relations, source versions and tool bindings. |
| Apply declared requirements | Unresolved support, contradictions and affected claims. |
| Compare controlled runs | Case-level differences linked to evidence changes. |
| Repair and recheck | Recorded repair, review owner and remaining failures. |

Source: Workshop synthesis / original teaching example

# Finish with an evaluation card

| Record | Decision it enables |
| --- | --- |
| Task, population and reference process | Know where the result applies. |
| Case families, controls and model revision | Reproduce and interpret the comparison. |
| Correctness, support, sensitivity and intervals | Assess behavior and uncertainty separately. |
| Known gaps, owner and retest trigger | Repair, limit or withhold deployment. |

Source: Choi, Stanford CS224N Lecture 11 (2026); Dror et al., Statistical Significance in NLP (ACL 2018)

# Five objectives: what can you now demonstrate?

| Objective | Evidence from this session |
|---|---|
| Define success | Write the task contract and abstention condition. |
| Choose measures | Separate correctness, support, stability and calibration. |
| Control the comparison | Specify paired changes, repeats and uncertainty. |
| Diagnose responsibly | Distinguish graph dependencies from internal causal tests. |
| Decide what follows | Name a repair, retest and bounded release claim. |

Source: Workshop synthesis / original teaching example

# Reading route: lectures and primary methods

- Choi, Stanford CS224N Lecture 11 (2026) — benchmark design, metrics and evaluation protocols.
- Ribeiro et al., CheckList (ACL 2020) — minimum, invariance and directional tests.
- DeYoung et al., ERASER (ACL 2020) — rationale support and perturbation diagnostics.
- Dror et al., Statistical Significance in NLP (ACL 2018) — selecting paired significance tests.
- Guo et al., Calibration (ICML 2017); Meng et al., Causal Tracing / ROME (NeurIPS 2022) — confidence and internal interventions.

Source: Workshop synthesis / original teaching example